

NAZIM HUSSAIN

AI Engineer · Fine-Tuner · LLM & RAG Specialist · Full Stack Developer

nazimali8172@gmail.com
[GitHub](#) →

0333-7006557
[LinkedIn](#) →

Sindh, Pakistan
[Portfolio](#) →

About Me

Final Year BS Artificial Intelligence student with GPA 3.2/4.0 at Quaid-e-Awam University of Engineering, Sciences & Technology, specializing in LLMs, RAG pipelines, agentic AI systems, and full-stack AI deployment. Built and shipped a multilingual legal RAG pipeline indexing 500+ legal documents with FAISS + BM25 hybrid retrieval and cross-encoder re-ranking achieving sub-2s response latency, a CNN digit recognizer at 92% accuracy on 60,000 training samples deployed live on Flask, an AI trading bot processing real-time OHLCV chart data via OpenCV with automated signal generation, and a Face Net-based face recognition system running at 30 FPS on edge hardware. Stack: Python, Lang Chain, Lang Graph, Fast API, TensorFlow, OpenCV, Docker production-used not just listed. Active Kaggle (9 badges). Seeking AI/ML internships and full-time roles.

Skills

AI / ML	Python · Machine Learning · Deep Learning · Transformers · PyTorch · TensorFlow · Scikit-learn · CNNs · RNNs · ANNs · NLP · LLMs · RAG · Prompt Engineering · Fine-tuning · MLOps
AI Tools	OpenAI · Hugging Face · LangChain · LangGraph · LlamaIndex · FAISS · BM25 · Pinecone · Chroma · OpenRouter · Gradio · Streamlit
Backend	Flask · FastAPI · Django · REST APIs · GraphQL · PostgreSQL · SQLite · MongoDB · Redis · JWT / OAuth · Docker · Docker Compose
Frontend	HTML · CSS · Tailwind CSS · JavaScript · TypeScript · React · Next.js · Shadcn/ui
Computer Vision	OpenCV · YOLOv8 · FaceNet · MediaPipe · Image Classification · Object Detection · Facial Recognition · Pose Estimation · Segmentation
Dev Tools	Git · GitHub · GitHub Actions · Docker · Vercel · Netlify · Postman · VS Code

Projects

Pakistan Legal Advisor

A multilingual (Urdu & English) legal assistant powered by a production RAG pipeline indexing 500+ Pakistani legal documents. Architecture: PDF ingestion → recursive text chunking → dual retrieval (FAISS dense vectors + BM25 sparse) → cross-encoder re-ranking → LLM answer generation with source citation. Achieves sub-2s end-to-end response latency. Deployed publicly on Hugging Face Spaces.

[Multilingual Urdu/English] [RAG Pipeline] [FAISS + BM25] [Re-ranking] [LLM Integration] [LangChain]

Digit Recognition System

CNN trained on 60,000 MNIST samples achieving 98% test accuracy. Applied dropout regularization (rate 0.5) and real-time data augmentation (rotation, zoom, shift) to reduce overfitting by ~4% vs baseline. Deployed via Flask with a canvas-based web UI for live digit prediction — averages <100ms inference per request.

[CNN] [MNIST] [92% Accuracy] [Flask] [TensorFlow] [Data Augmentation]

AI-Powered Trading Bot

Automated trading system that processes real-time OHLCV candlestick charts using OpenCV (Canny edge detection, contour analysis) to identify market structure breaks, volume spikes, and retest zones. Generates timestamped buy/sell signals with configurable risk parameters. Back tested across 6 months of historical data with 62% signal accuracy on trending markets.

[OpenCV] [Algorithmic Trading] [Python Automation] [MTG Strategy] [Back testing]

Face Recognition System

Real-time face detection and recognition pipeline using OpenCV (Haar cascades + DNN detector) and FaceNet 512-d embeddings with cosine similarity matching. Achieves 30 FPS on standard hardware and ~95% recognition accuracy across a 50-person database. Optimized for low-latency inference on edge devices with a sub-200ms identification pipeline.

[OpenCV] [FaceNet] [Real-time 30 FPS] [Edge Devices] [Cosine Similarity]

Experience

Jan 2022 — Jan 2023

Computer Operator - Citizen Police Liaison Committee (CPLC), SBA

- Managed and maintained computer systems and organizational databases, ensuring data integrity and system reliability
- Supported data entry, verification, and reporting workflows across departments
- Troubleshoot software and hardware issues, minimizing downtime and improving team efficiency

Education

Academic Background

2022 — 2026 (Expected)

BS Artificial Intelligence | GPA: 3.2 / 4.0 Quaid-e-Awam University of Engineering, Sciences & Technology

- Final-year student with strong academic focus on Machine Learning, Deep Learning, Computer Vision, and Natural Language Processing
- Advanced coursework: Neural Networks, Transformers, NLP, Computer Vision, MLOps, Optimization Techniques
- Built several end-to-end AI projects: multilingual RAG Legal Advisor, real-time Face Recognition System, CNN Digit Recognizer, AI Trading Bot
- Active Kaggle with 9 badges — consistently applying classroom concepts to real-world datasets
- Holds Coursera certifications: AI Essentials (2024) and Generative AI with Capstone Project (2024)

Contact

Let's Build Something

Actively seeking AI/ML internships, full-time opportunities, and impactful collaborations. Whether you're hiring, building, or exploring innovative ideas let's connect.

Email: nazimali8172@gmail.com | Phone: 0333-7006557